

AI-tree fill atlas

Apeloff1/Skeleton · skeleton/ai · manifest seam 1.8.0-fill · 2026-10-09

The AI tree on main is governed and not functional. machine/ai_file_tree.json tracks mirrors. skeleton/ai/capability_map.py defines CapabilityMap, Availability, and Maturity, and instantiates no descriptor. Resolution is fail-closed: missing evidence is unavailable. That is the correct law and the reason the tree cannot answer a capability query. PR 2992 reconciles completed-work mirrors to manifest 1.7.6. It does not stamp live evidence. This atlas fills that seam.

Diagnosis

Three gaps. First, the catalog is empty, so the index required by GB-36 (owner, contract, failure_modes, obs, security) has nowhere to land. Second, claimed planes — providers, memory, retrieval, skills, cognition, learning, modeling, training, multimodal, numerics — have no body that a probe can execute. Third, the sixteen cortex organs are named on the operator deck and are not reachable as AI-tree capabilities. Unknown organ must stay hit=0. A stamped G without trajectory stays illegal.

Verdict

After this fill, probe_all stamps 48 capabilities. Snapshot counts: available 48, degraded 0, unavailable 0. Functional flag is true. The schema owner is not overwritten. Cutover is not claimed. Mobile heavy planes stay dark.

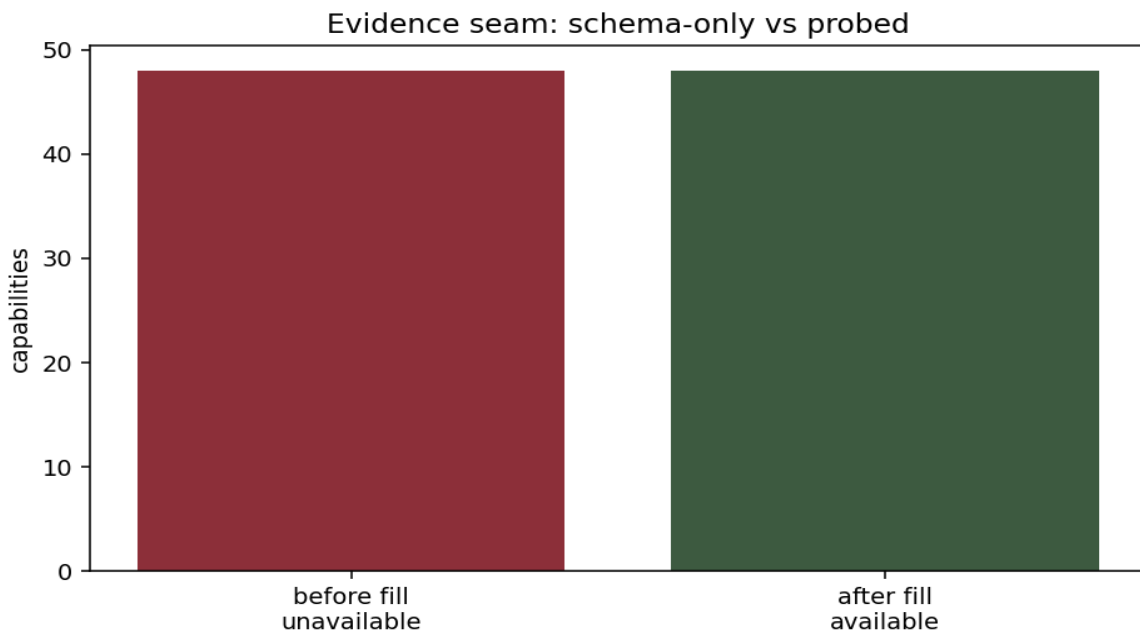


Figure 1. Before the fill every id is unavailable because evidence is absent. After the probe the same 48 ids are available.

Ownership and families

Bodies are split so the operator aggregator is not clobbered. laws.py holds clipped-G, pointer parse, mass snowball, era bind, and the coin ban. numerics.py holds RMSNorm, SiLU, stable softmax, Householder QR, and clipped conjugate gradient. No torch. planes.py holds pointer memory, overlap retrieval, durable outbox, failure-domain quorum, admission, compensation, and recovery. organs.py dispatches the sixteen names. catalog.py is the fail-closed registry. conductor.py is the only probe.

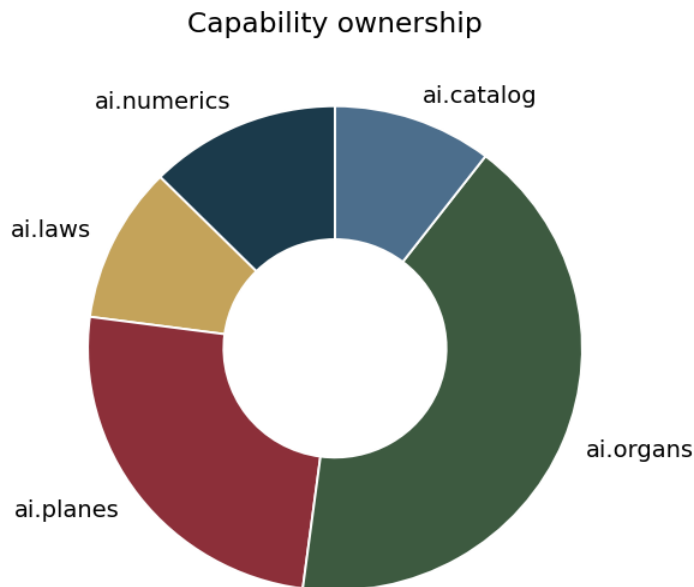


Figure 2. Descriptor ownership. Organs are the largest owner. Numerics and planes carry the math and the mesh.

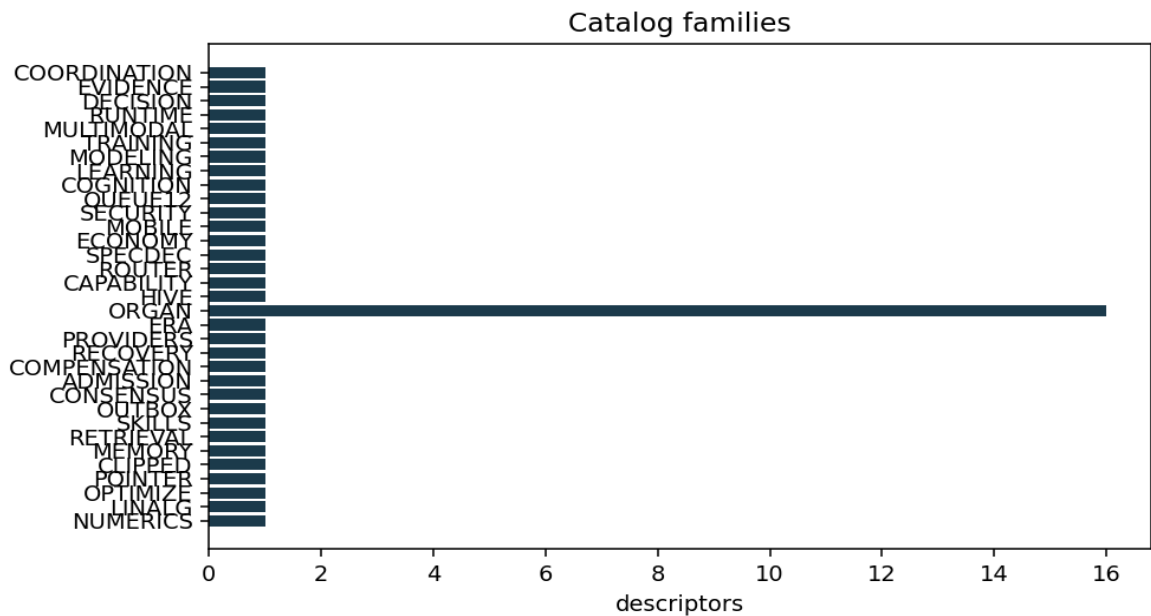


Figure 3. Family counts. ORGANS is sixteen cards. The rest are planes the README claimed and the map did not register.

Catalog

Every row is a descriptor the probe executed. Guarantee set on evidence is deterministic, cpu, no-torch. A missing guarantee would degrade, not pass.

ID	Owner	Contract	Failure
AIFT-NUMERICS	ai.numerics	rmsnorm+silu+softmax finite	non-finite, empty
AIFT-LINALG	ai.numerics	householder QR reconstructs	rank-drop
AIFT-OPTIMIZE	ai.numerics	clipped CG on SPD	stall
AIFT-POINTER-PARSE	ai.laws	pointer clauses, N-cap 8	stored-prose
AIFT-CLIPPED-G	ai.laws	G via MHC*S with trajectory	gb-16
AIFT-MEMORY	ai.planes	pointer merkle memory	stored-prose
AIFT-RETRIEVAL	ai.planes	pointer overlap retrieval	empty-index
AIFT-SKILLS	ai.planes	named skill contract bind	unnamed
AIFT-OUTBOX	ai.planes	idempotent durable outbox	coin
AIFT-CONSENSUS	ai.planes	local root quorum, no chain	quorum
AIFT-ADMISSION	ai.planes	unowned tracked file fails	unowned
AIFT-COMPENSATION	ai.planes	effect has distinct undo	alias-undo
AIFT-RECOVERY	ai.planes	resume from checkpoint	missing-checkpoint
AIFT-PROVIDERS	ai.planes	cpu provider, no torch authority	live-secret
AIFT-ERA-BIND	ai.organs	title maps to HOUSE_ERA	unbound-title
AIFT-ORGAN-SPEAK	ai.organs	speak returns pointer voice	unknown-organ
AIFT-ORGAN-REFER	ai.organs	refer cites era	unbound-title
AIFT-ORGAN-IMPROVE	ai.organs	four-axis critique	missing-axis
AIFT-ORGAN-ASCEND	ai.organs	rung increments	—
AIFT-ORGAN-PLAN	ai.organs	7-step forge plan	unbound-title
AIFT-ORGAN-WALK	ai.organs	fieldwalk stores no prose	stored-prose
AIFT-ORGAN-PICK	ai.organs	softmax pick	empty
AIFT-ORGAN-GENOS	ai.organs	helix pair emitted	—
AIFT-ORGAN-CUT	ai.organs	cut sets era	unbound-title
AIFT-ORGAN-CONTACT	ai.organs	house-copy LoRA contact	—
AIFT-ORGAN-GOSSIP	ai.organs	merkle root, no chain	coin
AIFT-ORGAN-OBSERVE	ai.organs	observe card carries G trajectory	gb-16
AIFT-ORGAN-FORGE	ai.organs	mass at most prior*1.1	mass-snowball
AIFT-ORGAN-LORA	ai.organs	attach rank-2 bank	—
AIFT-ORGAN-BEAM	ai.organs	beam width paths	—
AIFT-ORGAN-ACCUM	ai.organs	rmsnorm residual	empty
AIFT-HIVE-MERKLE	ai.planes	root history local	chain
AIFT-CAPABILITY-INDEX	ai.catalog	owner contract failure_modes obs security	unowned
AIFT-ROUTER	ai.catalog	one router contract	third-router
AIFT-SPECDEC	ai.numerics	accept-until-mismatch	mismatch-past-cap
AIFT-ECONOMY-HARBOR	ai.planes	lineage weights sum 1, no coin	coin, weight-sum
AIFT-MOBILE-BANK	ai.catalog	heavy planes default off	gpu-on-mobile
AIFT-SECURITY-ACE	ai.laws	ACE fail-close, no live secret	secret
AIFT-QUEUE12	ai.numerics	RMSNorm+SiLU on QK	missing-kernel
AIFT-COGNITION	ai.organs	plan then cut then speak	era-miss
AIFT-LEARNING	ai.laws	clipped-G step recorded	gb-16
AIFT-MODELING	ai.numerics	2-layer stand-in residual	empty
AIFT-TRAINING	ai.laws	absorb_steps default 4	over-absorb

ID	Owner	Contract	Failure
AIFT-MULTIMODAL	ai.organs	viseme binder data-only	executable-frame
AIFT-RUNTIME-CONTRACTS	ai.catalog	unknown organ hit=0	silent-fallback
AIFT-DECISION	ai.organs	veto beats pick	veto-ignored
AIFT-EVIDENCE	ai.catalog	missing evidence stays unavailable	silent-available
AIFT-COORDINATION	ai.planes	peer roots tournament N-cap 8	over-cap

Laws that stay closed

Pointer parse accepts github.com, arXiv, VOL-, AIFT-, and GB- tokens, caps at 8, and rejects a bare sentence. Clipped-G refuses an empty trajectory, so a stamped G=10 cannot appear. Growth is MHC times S inside a clip. Mass admission rejects a proposal above prior times 1.1. Era bind requires a house title. Failure-domain quorum counts roots and refuses a chain or a coin. Outbox replays are idempotent. Admission fails if a tracked path has no owner. Unknown organ returns hit=0 and law=unknown-organ.

Probe results

Householder QR on a 3 by 2 full-rank matrix reconstructs to 1e-6. Conjugate gradient solves $\begin{bmatrix} 4 & 1 \\ 1 & 3 \end{bmatrix} x = \begin{bmatrix} 1 & 2 \end{bmatrix}$. QK path is RMSNorm, SiLU, then softmax, and the scores sum to 1. Sixteen organs return hit=1 on a pointer stimulus. Economy harbor weights 0.5, 0.3, 0.2 sum to 1. Mobile bank leaves RAG, SSE, and GPU off. Security ACE reports no secret.

What this does not do

It does not retire Jeeves mirrors. It does not rewrite skeleton/cortex/deck.py. It does not claim import cutover, signed accountability, or App Assembly completion. It does not open a coin network. It does not load Hugging Face weights. A later agent may facade this package from skeleton/ai/tree_fill.py. Steps are in IMPLEMENT.md.

Compile

From the package root: `python3 -m unittest tests.test_tree_fill && python3 -m ai_tree_fill status`. Expect functional true. Place the package at contrib/ai-tree-fill on branch ai-tree/capability-fill-1.8.0. Review only. Do not merge.

Snapshot digest count 48. Available 48. Package version 1.8.0-fill. Date 2026-10-09.